

# Mechanics of displacive instabilities in solids

*Habilitation à Diriger des Recherches (HDR)*

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31 Mars 2026

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# Scientific Context: Nonlinearity & Instability

**The Challenge:** Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

## Key Phenomena:

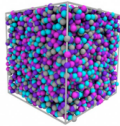
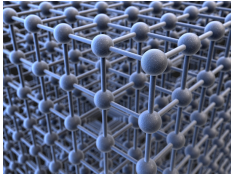
- ▶ **Phase Transitions:** Martensitic transformations.
- ▶ **Plasticity:** Dislocation-mediated flow and avalanches.
- ▶ **Fracture:** Brittle-to-ductile transitions and damage.

## Unifying Perspective

Treating these diverse phenomena as *displacive instabilities* driven by the competition between elastic energy and lattice symmetry.

# Non-linear phenomena

## Crystals (PDFs)



Solid-to-solid transitions in lattice structures.

## Meta-materials (PNGs)



Structural rearrangements in architected materials.